

TUZLA, Bosnia and Herzegovina

WMO: 145570

Lat: 44.5420N

Lon: 18.6851E

Elev: 302

StdP: 97.75

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.0	-14.0	1.2	-11.7	-10.8	1.5	-7.0	6.1	2.7	5.3	1.6	1.1	260	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	33.8	21.4	32.0	21.2	30.1	20.8	23.9	30.0	22.7	28.9	21.7	27.5	2.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.3	26.8	20.9	16.2	25.8	19.9	15.2	24.4	73.1	30.2	68.6	29.0	64.7	27.9	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.8	4.1	-17.0	36.7	4.8	2.3	-20.5	38.3	-23.3	39.7	-26.0	41.0	-29.5	42.7		
(5)			-17.1	25.8	4.8	1.8	-20.6	27.0	-23.4	28.1	-26.1	29.1	-29.6	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	0.0	2.5	7.0	11.6	16.3	20.0	21.3	21.4	16.5	12.5	6.9	1.4		
	DBStd	8.54	4.86	4.75	5.01	4.13	3.57	3.59	2.94	3.19	3.24	4.29	4.76	4.45		
	HDD10.0	1082	312	211	118	31	2	1	0	0	1	24	116	268		
	HDD18.3	2815	569	442	351	202	83	25	8	10	73	184	343	524		
	CDD10.0	1630	1	2	26	79	196	300	351	353	194	102	23	2		
	CDD18.3	321	0	0	0	1	19	75	100	105	17	4	0	0		
	CDH23.3	3829	0	0	8	64	319	851	1111	1161	243	71	0	0		
	CDH26.7	1498	0	0	1	9	87	331	481	524	57	9	0	0		
(14) Wind	WSAvg	1.4	1.3	1.5	1.8	1.8	1.6	1.4	1.4	1.3	1.3	1.2	1.4	1.4		
(15) Precipitation	PrecAvg	904	61	56	50	77	92	105	104	79	71	54	77	78		
	PrecMax	1232	132	120	135	137	224	177	290	213	151	137	140	158		
	PrecMin	600	10	13	8	19	23	45	14	15	21	0	13	4		
	PrecStd	125	33	30	34	28	52	41	59	49	30	36	29	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.8	18.9	24.2	28.0	31.1	34.0	36.0	37.9	30.8	27.9	21.9	17.2		
		MCWB	10.9	10.5	13.0	16.2	20.1	21.1	20.6	20.9	19.9	18.2	14.6	11.2		
	2%	DB	12.9	16.0	21.2	25.1	29.1	32.1	33.6	34.1	28.1	25.2	19.8	13.7		
		MCWB	8.7	9.3	11.8	15.1	19.0	21.1	21.4	21.3	19.0	16.6	13.1	9.4		
	5%	DB	9.2	13.1	19.0	22.9	27.0	30.8	31.6	31.9	26.2	23.0	17.8	10.8		
		MCWB	6.8	8.0	11.2	13.7	18.0	21.2	21.4	21.4	18.3	16.2	12.3	8.0		
	10%	DB	6.2	10.1	16.8	20.2	24.8	28.7	29.2	29.8	24.1	20.4	15.2	8.1		
		MCWB	5.0	6.5	10.6	12.9	17.3	20.3	20.7	20.9	17.5	15.1	11.2	6.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	11.5	15.9	18.4	22.6	24.6	25.1	25.4	22.6	19.5	15.5	11.6		
		MCDB	15.6	16.4	19.9	24.1	27.4	30.2	29.9	31.2	27.0	24.3	19.6	15.7		
	2%	WB	9.1	9.9	13.1	16.3	20.2	23.1	23.6	23.8	20.4	17.9	14.0	10.0		
		MCDB	12.2	15.2	18.8	22.2	26.2	28.8	29.7	30.5	25.9	23.4	18.1	13.0		
	5%	WB	7.1	8.5	12.0	15.1	18.9	21.8	22.3	22.7	19.2	16.8	12.5	8.3		
		MCDB	8.9	12.6	17.5	20.7	24.9	27.8	28.7	28.8	23.8	21.6	16.9	10.4		
	10%	WB	5.1	6.9	10.8	13.9	17.6	20.9	21.2	21.6	18.0	15.6	11.2	6.6		
		MCDB	6.4	9.7	16.2	18.7	23.0	27.3	27.6	27.0	22.6	19.5	15.3	7.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	10.2	12.6	12.8	13.1	13.0	13.3	13.9	12.6	13.2	10.1	8.1		
		MCDBR	13.9	16.4	17.7	18.2	17.9	16.9	17.8	18.9	17.2	17.5	15.1	13.5		
		MCWBR	9.8	10.5	9.5	9.3	8.8	7.6	7.4	8.0	8.7	9.6	9.2	9.7		
	5% WB	MCDBR	12.9	15.0	15.2	15.3	15.9	15.2	15.4	15.2	15.0	15.4	13.4	11.9		
		MCWBR	9.5	10.0	9.0	8.8	8.9	7.7	8.0	8.2	8.3	8.8	8.5	9.0		
(40) Clear-Sky Solar Irradiance	taub	0.332	0.361	0.397	0.453	0.444	0.440	0.442	0.442	0.442	0.424	0.388	0.349	0.330		
	taud	2.375	2.271	2.221	2.084	2.159	2.219	2.214	2.212	2.212	2.236	2.332	2.396	2.385		
	Ebn at Noon	774	811	829	809	830	834	827	811	792	771	747	735	70		
	Edh at Noon	78	103	123	153	147	139	138	133	120	96	76	70	70		
(44) All-Sky Solar Radiation	RadAvg	1.31	2.02	3.22	4.29	5.30	5.89	6.08	5.45	3.76	2.62	1.58	1.08			
	RadStd	0.20	0.27	0.41	0.51	0.50	0.54	0.48	0.59	0.44	0.31	0.15	0.18			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		+0.66	N/A	N/A	N/A	+0.37	N/A	-117	-184	+116	N/A

Nomenclature: See separate page